Chapter 24

Formula Answer

1	2	3	4	5	6	7	8	9	10

1) At Tyler's Pizza Palace, a large pizza with no toppings (sausage, pepperoni, meatball, mushroom, onion, and others) costs \$14.00. If Tyler's charges \$$t$ for each topping on their large pizza, **what is the cost for a large pizza with g toppings**?

A) $\$(14.00\ x\ (gt))$

B) $\$(14.00 + gt)$

C) $\$(28.00 + gt)$

D) $\$(14.00 + t/g)$

E) $\$(14.00 + g/t)$

2) If the price of g gallons of gasoline costs d dollars. **How much would it cost** to fill a tank with gasoline that holds a capacity of c gallons assuming the tank was empty?

A) $\frac{c}{dg}$ B) $\frac{cg}{d}$ C) $\frac{cd}{g}$ D) $\frac{d}{cg}$ E) $\frac{c}{cdg}$

3) How many inches are there in ***y* yards *f* feet and *i* inches**?

(*Note; there are 36 inches in a yard and 12 inches in a foot*)

A) $432yfi$

B) $36y + 12f + i$

C) $y + 36i + 12\ f$

D) $1.36\ y + 1.12f + i$

E) $y + f + i$

4) If a car travels m miles in h hours, how many miles at the same rate will the car travel in $\boldsymbol{h+1}$ **hours**?

A) $\frac{m}{h(h+1)}$

B) $\frac{h}{m(h+1)}$

C) $\frac{(h-1)}{m(h+1)}$

D) $\frac{m(h+1)}{h}$

E) $\frac{(m-1)}{m(h+1)}$

5) $\boldsymbol{R}$ students' typically read at an average rate of $\boldsymbol{p}$ pages per hour. If $\boldsymbol{s}$ of the students were to take a special reading booster pill to make them able to read $\boldsymbol{t}$ times as fast as usual, **which expression would yield the total** number of pages read by the $\boldsymbol{R}$ group of students in $\boldsymbol{h}$ hours?

A) $(R-s) \bullet \{(sp) - t) + (hR)\}$

B) $h \bullet \{p(R-s) + spt\}$

C) $sp \bullet \{t(R-s) + hp\}$

D) $h \bullet \{p(t-s) + spR\}$

E) $s \bullet \{p(s-Rh) + pt\}$

6) If water drains from a tank at a constant rate of $\boldsymbol{r}$ liters every $\boldsymbol{s}$ seconds and the capacity of the tank is $\boldsymbol{c}$ liters, how many seconds will it take for the tank to drain completely?

A) $\frac{rs}{c}$ B) $\frac{2rc}{s}$ C) rsc D) $\frac{cs}{r}$ E) $2rs$

1	2	3	4	5	6	7	8	9	10

7) In the previous question, **how many seconds** will it take for the tank **to lose p percent of its water**?

A) $\frac{rs}{c}$ B) $\frac{ps}{100rc}$ C) $\frac{ps}{100c}$ D) $\frac{pcs}{100r}$ E) $\frac{rs}{c}\ x\ 100$

8) In a particular game where $A > B > C$, players score either A, B, or C points per turn but only receive one score per turn. With only positive scoring, if A points is d % more points than B points and three times as many points as C points, what is the maximum number of points a player can receive in 10 turns if it is never possible to receive the same score twice in a row and every player must score only C points on every turn where the turn number is divisible by 3?

A) $3A + [AC / 100(1 + d)]$

B) $3[A(100/ (100 + d)) + 15C]$

C) $3\{A + 2B + C[100/ (100 + d)]\}$

D) $4[(100 + d)/ 100] + 3B + 3C$

E) $3(B + C) + [4B(100 + d)/ 100]$

Extremely Difficult

Level 10

9) In the made-up game from the previous question, with all the same rules in place, what percentage would the maximum score increase if a player could score the same score twice in a row?

A) $\left(\frac{2B[(100+d)/100]}{4A+3B+3C}\right) \cdot 100\ \%$

B) $\left(\frac{2\{B[(100+d)/d]-B\}}{4A+3B+3C}\right) \cdot 100\ \%$

C) $\left(\frac{2\{B[(100+d)/d]-A\}}{4A+3B+3C}\right) \cdot 100\ \%$

D) $\left(\frac{3\{B[(100+d)/d]-A\}}{4A+3B+3C}\right) \cdot 100\ \%$

E) $\left(\frac{3\{B[(100+d)/100]-B\}}{4A+3B+3C}\right) \cdot 100\ \%$

Extremely Difficult

Level 10

10) If **train A** travels at ***r* miles in *s* hours** and **train B** travels at ***j* miles in *k* hours**, which of the following would be the positive difference between the average speeds in miles per hour of the two trains?

A) $|(r(j/k) - s(j/k)|$
B) $|(r/s) - jk|$
C) $|(r/s) - (j/k)|$
D) $|(j/k) - rs|$
E) $|((j/k)(r/s)|$

11) If a particular cookie recipe requires ***g*** grams of sugar and ***c*** ounces of milk per serving, which of the following is the amount of sugar and milk needed respectively for ***s*** servings?

A) (g/s) grams of sugar and (c/s) ounces of milk
B) (gs) grams of sugar and (cs) ounces of milk
C) $(4gs)$ grams of sugar and (c/s) ounces of milk
D) (g/c) grams of sugar and (c/s) ounces of milk
E) (gc) grams of sugar and (cs) ounces of milk

12) Mitchell is trying to figure out his car's gas mileage the old-fashioned way. He checked his odometer reading after filling his tank and logged the mileage ***m***. A few days later, Mitchell filled his car's gas tank again, noting the new mileage ***n*** on the odometer. Finally, Mitchell logged the amount of gas ***g*** in gallons his car needed to be full of gas again. How would Mitchell calculate his car's gas mileage in miles per gallon?

A) $\frac{(n-m)}{g}$

B) $\frac{(m-n)}{g}$

C) $g(m-n)$

D) $g(n-m)$

E) $\frac{(nm)}{g}$

13) To earn an "A" in her Math class, Jess must have at least a 90% average on three equally weighted tests. If Jess scored a % on test 1 and b % on test 2, which of the following represents the minimum Jess must score on her third and final test to **get an "A" in math**?

A) $(270 - (a + b))\%$

B) $\left(\frac{a + b + 90}{3}\right)\%$

C) $\left(\frac{2a+2b}{90}\right)\%$

D) $\left(\frac{2a + 2b}{270}\right)\%$

E) $(90 - (a + b))\%$

14) Naveen and his ***f*** friends went on a ***d***-day fishing trip, and the entire group caught a total of ***s*** fish. Which expression below represents the average number of fish caught per day by each person?

A) $\frac{\mathrm{d(f-1)}}{s}$

B) $\frac{1}{d}\left(\frac{s}{f+1}\right)$

C) $\frac{s(f+1)}{d}$

D) $\frac{fs}{d}$

E) $\frac{ds}{f}$

15) If Golander gets paid ***d*** dollars per week when working 40 hours per week, which expression below represents how much Golander gets paid by the hour?

A) $\$140d$ B) $\$\ \frac{d}{40}$ C) $\$\frac{40}{d}$ D) $\$40(d^2)$ E) $\$160d$

1	2	3	4	5	6	7	8	9	10

16) It takes a minimum of $\boldsymbol{s}$ students to fill $\boldsymbol{b}$ busses with the same number of seats on each bus. If $\boldsymbol{r}$ seats are removed from each bus, how many students will it take to fill $\boldsymbol{b+2}$ busses?

A) $\frac{2b}{(r+s)}$

B) $\frac{2b}{(r-s)}$

C) $(b+2)\left(\frac{b}{s}-r\right)$

D) $br+2s$

E) $(b+2)\left(\frac{s}{b}-r\right)$

Solutions Chapter 24

Formula Answer

1) B	5) B	9) E	13) A
2) C	6) D	10) C	14) B
3) B	7) D	11) A	15) B
4) D	8) E	12) A	16) E

(Formula Answer is a made-up name for questions where the answers look like Formulas)

The best way to approach a "**formula answer**" type question is to make up numbers that work well and then plug them into the answer choices to find the correct solution. Numbers that work well divide evenly into each other and make sense. For instance, if a question asks for m miles per g gallons, a good choice might be 10 miles per 2 gallons to give a simple unit price of 5 miles per gallon as opposed to say 13 miles in 4 gallons which would result in a unit price of 3.25 miles per gallon. It is never a good idea to choose the number 1 as a number to substitute. Students can identify most "**formula answer**" questions by just looking at the possible solutions. If the answers look like formulas with variables, it is usually a **formula answer** question. If more than one answer works using the plug-in method, substitute different numbers. Students may recognize that the correct answer choice mirrors the mathematical steps taken to arrive at the plug-in solution.
Note: *Some **formula answer** questions can be quickly answered by looking at the solutions, but students can solve most efficiently using the plug-in method.*

Formula answer questions are easy to identify because they are usually wordy.

The plug-in method will be applied to answer all the questions in this section.

1) First, choose a value for t and g that are easy to work with except the number 1. For example, say t, the number of dollars per topping is equal to 2, and say g, the number of toppings = 3. The base cost for the pizza is \$14.00, plus 3 toppings at 2 dollars each (3 • \$2.00 = \$6.00) for a total of \$14.00 + \$6.00 = \$20.00. Plugging in

2 for t and 3 for g, the only answer choice that is also equivalent to \$20 is answer choice B**;** \$(14.00 + gt); (14.00 + (3)(2)) = 20 or \$20.00

\$14.00 + gt Answer B Level 6

2) **Make up values** for g, d, and c that are easy to work with but never use the number 1. Let's say (g) = (5) gallons of gasoline costs (d) = (\$20). *Always calculate the "per unit value" (e.g., price per gallon, miles per hour, number of sodas per case)***.** In this question, 5 gallons of gasoline for \$20 is 20/5 = \$4 per gallon. Choose c = 3 for the number of gallons the tank holds, which means it would cost \$4 • 3 = \$12 to fill the empty tank. Plug in; g = 5, d = 20, and c =3 into the possible answer choices given; the correct answer should equal \$12.

Answer choice C works, $\frac{\text{cd}}{\text{g}} = \frac{(3)\bullet(20)}{5}$ = \$12

$\frac{\text{cd}}{\text{g}}$ Answer C Level 7

3) **Make up values for the variables**; y, f, and i that are easy to work with but never use the number 1. Let's say y = 2, f = 3, and i = 4. After making up the variables, the question becomes how many inches are there in 2 yards, 3 feet, and 4 inches; 2 yards = $2\ x\ 36 = 72$ inches, 3 feet = $3\ x\ 12 = 36$ inches and 4 inches; $72 + 36 + 4$ = 112 inches. Finally, check the answer choices; A) 432yfi; 432(2)(3)(4) will not equal 112, B) $36y + 12f + i$; 36(2) +12(3) + 4 = 112. That works, answer choice B. Notice, the same calculation was employed to arrive at the plug-in solution; 36 times the number of yards plus 12 times the number of feet plus the number of inches.

$36y + 12f + i$ Answer B Difficulty Level 7

4) **Make up values for the variables**; Let's say the car travels ***m* = 300** miles in ***h* = 5 hours** (calculate the unit speed) 300/5 = 60 miles per hour. How far will the car travel in h + 1 or 5 + 1 = 6 hours? The car would travel at 60 miles per hour for 6 hours; 60 • 6 = **360 miles**. Plug in m = 300 and h = 5 into each answer choice.

$\frac{m(h+1)}{h} = \frac{300(5+1)}{5} = \frac{300(6)}{5} =$ **360**

$\frac{m(h+1)}{h}$ Answer D Difficulty Level 9

5) **Choose *R, p, s, t,* and h** that make it easy to work with. *Never choose the number 1 for the plug-in technique.* Choose R = 10 students, read at an average rate of p = 20 pages per hour, where s = 3 students take a special reading booster pill, which makes them read t = 3 times faster. How many pages will the students read in h = 4 hours?

Breaking this down, 7 students read at an average rate of 20 pages per hour for 4 hours = 7 • 20 • 4 = 560 pages total and 3 students read at an average rate of 60 pages per hour for 4 hours = 3 • 60 • 4 = 720 pages total. The group of 10 students read a total of 560 + 720 = 1280 pages. Plug in R = 10, p = 20, s = 3, t = 3, and h = 4 and the correct answer will be the one that is equal to 1280.

$h \bullet \{p(R-s)+spt\}$ =

4 • $\{20(10-3)+(3)(20)(3)\}$ = 4 • $\{20(7)+(180)\}$ = 1280

$h \bullet \{p(R-s)+spt\}$ Answer B Difficulty Level 9.5

6) **Make-up values** for r, s, and c that are easy to maneuver. *Never choose the number 1 for the plug-in technique.* If the water drains at a constant rate of r = 10 liters every s = 5 seconds (calculate the liters per second speed) $^{10}/_{5}$ = 2 liters per second. The capacity of the tank; choose c = 20 liters so that it would take $^{20}/_{2}$ = 10 seconds to drain all the water.

Finally, plug in r = 10, s = 5, and c = 20 to each answer choice and find the correct choice that equals 10. Since $\frac{cs}{r} = \frac{20 \bullet 5}{10}$ = 10.

$\frac{cs}{r}$ Answer D Difficulty Level 8

7) How long does it take the tank to drain? Choose p = 20 percent of its water and keep all the other variables the same as in the previous question;

r = 10, s = 5, and c = 20. Since the capacity is 20 liter's, find how long it takes to lose 20 percent or .20 • 20 = 4 liters of water? At 2 liters per second, it will take 2 seconds to drain 4 liters or 20% of the water in the tank. The correct answer after plugging in the chosen values will be 2 seconds. Checking all the answer choices; $\frac{pcs}{100r} = \frac{20 \bullet 20 \bullet 5}{100 \bullet 10} = \frac{2000}{1000} = 2$ is the correct answer choice.

$\frac{pcs}{100r}$ Answer D Difficulty Level 9.5

8) The key to solving this problem is to choose the appropriate numbers. *Never select the number 1 for the plug-in technique.* Once the correct numbers are chosen, players must figure out how the game works. Since

$A > B > C$ and A is d % larger than B and three times as large as C, choose;

$A = 18$, $B = 10$, $C = 6$, $D = 80$.

(To check; 80% of 10 is 8, so 10 increased by 80% is 8 + 10 = 18 = A and

C = ($^1/_3$ of 18) = 6)

To maximize the game's score, score A points on every turn, and to minimize the score of the game, score C points on every turn.

1	2	3	4	5	6	7	8	9	10
A	B	C	A	B	C	A	B	C	A
18 points	10 points	6 points	18 points	10 points	6 points	18 points	10 points	6 points	18 points

Since for every turn number divisible by three, you can only score C points, turns three, six, and nine must score C or 6 points. Since you cannot score the same score twice in a row, turns 1 and 2, 4 and 5, and turns 7 and 8 must all be either score A then B or B then A. In round 10, you would score A points to maximize your score. With the chosen variables the maximum score is displayed in the table above. The total score would = $18 + 10 + 6 + 18 + 10 + 6 + 18 + 10 + 6 + 18 = 120$. Finally, plug in the chosen variables and the selection that equals 120 is the correct answer. $3(B + C) + [4B (100 + d)/ 100] =$

$3(10 + 6) + [4 \bullet 10(100 + 80)/ 100] = 3(16) + [40(9)/ 5] = 48 + 72 = 120$

3(B + C) + [4B (100 + d)/ 100] Answer E Difficulty Level **10**

9) Keeping all the scoring the same, if players could score the same score twice in a row, their new maximum score for ten rounds is shown below.

1	2	3	4	5	6	7	8	9	10
A	A	C	A	A	C	A	A	C	A
18 points	18 points	6 points	18 points	18 points	6 points	18 points	18 points	6 points	18 points

The new total score is 144 points which is an increase from 120 or an increase of 24 or a $^{24}/_{120}$ = .2 which is a 20% increase.

$\left(\frac{3\{B[(100+d)/100]-B\}}{4A+3B+3C}\right) \bullet 100\ \% = \left(\frac{3\{10[(100+80)/100]-10\}}{4\bullet18+3\bullet10+3\bullet6}\right) \bullet 100\ \%$

= .2 • 100% =20%

$\left(\frac{3\{B[(100+d)/100]-B\}}{4A+3B+3C}\right) x\ 100\%$ **Answer E** Difficulty Level **10**

10) Make up and substitute numbers that divide evenly into each other to avoid working with decimals. *Never choose the number 1 for the plug-in technique.* Making up numbers that work well; train A, $r = 30$ miles in $s = 2$ hours, so train A travels at $30/2\ =\ 15$ miles per hour and train B travels $j = 50$ miles in $k = 5$ hours or $50/5 = 10$ miles per hour. The positive difference with these made-up numbers would be$|10 - 5| = 5$ miles per hour. Finally, check the answer choices. A; $\left|30\left(\frac{50}{5}\right) - 2\left(\frac{50}{5}\right)\right|\ = |300 - 20| = 80$, B; $\left|\left(\frac{30}{2}\right) - 50(5)\right| = 235,$

C; $\left|\left(\frac{30}{2}\right) - \left(\frac{50}{10}\right)\right| = 10$. That is the correct solution. The correct answer is C, but not so fast. Always check all the answer choices in case more than one is correct. If this rare event occurs, test the correct answers with newly made-up numbers that are easy to use.

Plugging in to check for D;$\left|\left(\frac{50}{5}\right) - 30(2)\right| = 50$ and E;$\left|\left(\frac{50}{5}\right) x \left(\frac{30}{2}\right)\right|$=150.

$|(r/s) - (j/k)|$ Answer C Difficulty Level 7

11) Making up numbers and plugging them in is the easiest way to answer the not so easy formula answer type questions. Let's make up $g = 4\, grams,\ c = 8\, ounces,$ and $s = 2\, servings.$ *Never choose the number 1 for the plug-in technique.* For two servings, 4 grams of sugar is needed and 8 ounces of milk. How much is this per serving (per 1 serving)? Divide each quantity by 2; for one serving, 2 grams of sugar and 4 ounces of milk would be needed with these made-up numbers. Finally, plug in the made-up numbers into each answer choice to see which one matches the correct answer. The recipe calls for 2 grams of sugar and 4 ounces of milk. For answer choice A) (4/2) grams of sugar and (8/2) ounces of milk. That reduces to 2 grams of sugar and 4 ounces of milk. That is the correct answer. Now, check the rest of the answer choices in case there is a duplicate; B) (4*2) grams of sugar and (8*2) ounces of milk is not correct. Answer choices C, D, and E, do not work as well. There is no need to try all of them because it is easy to see that some have no chance of being the correct answer.

$\underline{(g/s)\ grams\ of\ sugar\ and\ (c/s)\ ounces\ of\ milk}$

Answer A Difficulty Level 6

12) Figuring out the gas mileage of an automobile in the 1980s was usually an unpleasant task for most. Now the car's computer does it for us. To correctly answer this question, start by making up numbers that work efficiently. Mitchell began with $m = 1000$ miles driven on his car with a full tank of gas. When Mitchell filled his car with gas a few days later, he looked at his odometer (The device that shows how many total miles an automobile has traveled), and it read $n = 1200\, miles$. To refill Mitchell's tank, it took exactly 25 gallons of gas. How many miles per gallon did Mitchell's car get. Now, figure out the answer logically. If Mitchell started with 1000 miles on the odometer and currently has 1200 miles on the odometer, he drove 200 miles. To find miles/gallon, divide the number of miles driven by the number of gallons used. Mitchell drove 200 miles/ 25 gallons or 8 miles/ gallon. Finally, plug into each answer choice to find the correct answer; A) $\frac{1200-1000}{25}$ = 8. That looks like the answer. Look at the other four answer choices just in case another one also equals 8. B would be negative, C is also negative, D is way too big, and E is way too big. It is easy to see only answer choice A works,

and the same calculation was used; $\frac{Ending\ Mileage - Beginning\ Mileage}{Gallons\ of\ gas\ used}$ to correctly answer the question when initially plugging in.

$\frac{n-m}{g}$ Answer A Difficulty Level 7.5

13) This question can be correctly answered by substituting made-up numbers. Be careful when choosing numbers by trying not to use numbers resulting in a fraction, as it will only complicate the question. Making up numbers, Jess scored *a* = 80% on test 1 and *b* = 90% on test 2. Now, think logically to find the correct answer with the made-up numbers. If Jess has three tests that count the same and scored an 80% and a 90%, she would need 100% on the final test to average 90%. Or, algebraically; $\left(\frac{80+90+x}{3} = 90\right)$ = $\left(\frac{170+x}{3} = 90\right)$, multiply both sides of the equation by 3, $170 + x = 270$ and $x = 100$. Therefore, the answer with those made-up numbers would be **100%**. Finally, plug the made-up numbers into each answer choice to find the correct answer;

A) $\big(270 - (80 + 90)\big)\%$ = **100%**
B) $\left(\frac{80+90+90}{3}\right)\% = \left(\frac{260}{3}\right)\% = 86.7\%$
C) $\left(\frac{2(80)+2(90)}{90}\right)\% = \left(\frac{340}{90}\right)\% = 3.78\%$
D) $\left(\frac{2(80)+2(90)}{270}\right)\% = \left(\frac{340}{270}\right)\% = 1.26\%$
E) $\big(90 - (80 + 90)\big)\% = -80\%$

Only answer choice A works correctly for the made-up variables. Note: If *a* = 90 and *b* = 90, answer choice B would also work. Make sure to check all the answer choices. If more than one answer choice works, choose different numbers. Also, look at the answer choices and select the answer that matches the original calculation performed in the plug-in step; $270 - (a + b)\%$;

$\big(270 - (A + B)\big)\%$ Answer A Difficulty Level 9

14) Sometimes, like in this question. making up numbers that are easy to use may be difficult. Many times, the numbers chosen can be altered to make the

question easier. For this question, make up random numbers to substitute for the variables, then adjust accordingly; f = 3 friends that went on a d = 5-day fishing trip, and the group caught a total of s = 30 fish. Be careful because this is somewhat of a trick question. There is a total of 4 people on the trip (Naveen plus three friends). So, four people went on a 5-day fishing trip. If a total of 30 fish were caught, each person caught an average of $\frac{30\text{ fish}}{4\text{ people}} = 7.5$ fish. Change the number of total fish to s = 32 since this would divide equally. Now each person caught $\frac{32\text{ fish}}{4\text{ people}} = 8$ fish. Since the crew went on the trip for five days, they caught $\frac{8\text{ fish each}}{5\text{ days}} = 1.6$ fish each per day. To avoid decimals, change the number of days of the trip to d = 4, which would result in $\frac{8\text{ fish each}}{4\text{ days}} =$ 2 fish each per day. Finally, with the new numbers substituted; f = 3, d = 4, and s = 32, the answer is two fish each per day. Check to see which answer choice is correct;

A) $\frac{d(f-1)}{s} = \frac{4(3-1)}{32} = \frac{8}{32} = \frac{1}{4}$

B) $\frac{1}{d}\left(\frac{s}{f+1}\right) = \frac{1}{4}\left(\frac{32}{3+1}\right) = \frac{32}{16} = \mathbf{2}$

C) $\frac{s(f+1)}{d} = \frac{32(3+1)}{4} = 32$

D) $\frac{fs}{d} = \frac{3*32}{4} = \frac{96}{4} = 24$

E) $\frac{ds}{f} = \frac{4*32}{3} = \frac{128}{3} = 42.67$

$\frac{1}{d}\left(\frac{s}{f+1}\right)$ Answer B Difficulty Level 9

15) To answer this question correctly, make up a value for d. If $d = 400$, then Golander earns \$400 per week for working 40 hours per week. Therefore, Golander would make $\frac{400}{40} = \$10/hour$. Finally, substitute into the answer choices;

A) $\$140d = \$140 * (400) = \$56{,}000$

B) $\$\frac{d}{40} = \$\frac{400}{40} = \$10$

C) $\$\frac{40}{d} = \frac{40}{400} = \frac{1}{10}$

D) $\$40(d^2) = \$40(400^2) = \$6{,}400{,}000$

E) $\$160d = \$160 * 400 = \$64{,}000$

$\$\frac{d}{40}$ Answer B Difficulty Level 6

16) Start by making up numbers that are easy to work with. If s = 20 students are to fill b = 2 busses with the same number on each bus, that would mean there are ten students on each bus. If $r = 2$ seats were removed from each bus, that would make each bus have 8 seats. If $b = 2$, then how many students will it take to fill b + 2 = 4 busses? Since each bus has 8 seats, it would take $8 * 4 = 32$ students to fill the 4 busses. Now check which answer choice is correct by calculating which one gives the correct answer of **32** with the made-up numbers substituted into the variables.

A) $\frac{2b}{(r+s)} = \frac{2(2)}{(2+20)} = \frac{4}{22} = \frac{2}{11}$

B) $\frac{2b}{(r-s)} = \frac{2(2)}{(2-20)} = \frac{4}{-18} = -\frac{2}{9}$

C) $(b+2)\left(\frac{b}{s} - r\right) = (2+2)\left(\frac{2}{20} - 2\right) = 4\left(\frac{-19}{10}\right) = -3.6$

D) $br + 2s = 2(2) + 2(20) = 44$

E) $(b+2)\left(\frac{s}{b} - r\right) = (2+2)\left(\frac{20}{2} - 2\right) = \mathbf{32}$

$(b+2)\left(\frac{s}{b} - r\right)$ Answer E Difficulty Level 9.5